Research Methodology

Module 4

Statistical Tools for Research

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1 Classification and Tabulation of Data

1.1 Classification

Classification is the process of **arranging raw data into groups or classes** according to common characteristics, so that it becomes simple and meaningful.

Bases of classification:

- **Qualitative (descriptive):** by attributes such as gender, religion, or course.
- **Quantitative (numerical):** by measurable values such as marks, age, or income.
- **Geographical:** by place, such as state or district.
- **Chronological (temporal):** by time, such as year or month.

Frequency Distribution

A frequency distribution shows **how many times** each value or class occurs.

- **Class interval:** a range of values, such as 10–20.
- **Class limits:** the lowest and highest values in a class.
- **Class width:** upper limit minus lower limit.
- **Mid-point:** (lower limit + upper limit) / 2.
- **Cumulative frequency:** running total of frequencies.

Marks	Frequency (f)	Mid-point (m)	Cumulative f
0–10	4	5	4
10–20	8	15	12
20–30	15	25	27
30–40	9	35	36
40–50	4	45	40
Total	40		

> **Note** The number of classes is usually kept between **5 and 15**. Classes should be **exhaustive** (cover all data) and **mutually exclusive** (no value belongs to two classes).

1.2 Tabulation

Tabulation is the **systematic arrangement of data in rows and columns**, so that it can be compared and analysed easily.

Parts of a table: table number, title, caption (column headings), stub (row headings), body, head-note, and footnote or source.

- **One-way (simple) table:** one characteristic only.
- **Two-way table:** two characteristics together (e.g. course by gender).
- **Higher-order table:** three or more characteristics.

> **Exam Tip** Classification vs. tabulation: **classification** groups the data; **tabulation** presents the grouped data in a table. Classification comes first.

2 Measures of Central Tendency and Dispersion

2.1 Central Tendency

A single value that represents the **centre** of the data.

Arithmetic Mean

$$\bar{x} = \frac{\sum x}{n} \qquad \text{(grouped data: } \bar{x} = \frac{\sum fm}{\sum f}\text{)}$$

- Uses all values, so it is the most widely used.
- Affected by extreme values (outliers).

Median

The **middle value** when data is arranged in order. For n values: the $\frac{n+1}{2}$-th value (odd n), or the average of the two middle values (even n).

- Not affected by outliers; best for skewed data such as income.

Mode

The **most frequent value**. A data set can have no mode, one mode, or more than one.

- Only measure usable for **nominal** data (e.g. most common course).

	Mean	Median	Mode
Meaning	Average	Middle value	Most frequent
Outliers	Strongly affected	Not affected	Not affected
Best for	Symmetric data	Skewed data	Categories
Data level	Interval, ratio	Ordinal and above	Any

> **Note** For a moderately skewed distribution: **Mode** $\approx 3 \times$ **Median** $- 2 \times$ **Mean**. In a perfectly symmetric distribution, mean = median = mode.

2.2 Dispersion

Dispersion measures **how spread out** the data is. Two data sets can have the same mean but very different spread.

Measure	Formula	Remark
Range	Maximum − Minimum	Simple, but uses only two values
Quartile deviation	$\frac{Q_3 - Q_1}{2}$	Ignores extremes
Mean deviation	$\frac{\sum \lvert x - \bar{x} \rvert}{n}$	Average absolute distance from mean
Variance	$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$	Average squared deviation
Standard deviation	$s = \sqrt{s^2}$	Most used; same unit as data
Coefficient of variation	$\frac{s}{\bar{x}} \times 100$	Compares spread of different data sets

For a whole population, divide by N instead of $n-1$ and use σ and μ.

Worked Example

Data: 4, 6, 8, 10, 12.

- Mean $= 40/5 = 8$. Median $= 8$. No mode.
- Deviations: $-4, -2, 0, 2, 4$. Squares: 16, 4, 0, 4, 16. Sum $= 40$.
- Sample variance $= 40/4 = 10$; sample SD $= \sqrt{10} \approx 3.16$.
- Population variance $= 40/5 = 8$; SD ≈ 2.83.
- Range $= 12 - 4 = 8$.

> **Exam Tip** Remember: **sample** uses $n-1$ (this corrects the underestimation of spread), **population** uses N. Mark this difference in exams.

3 Correlation and Regression Basics

3.1 Correlation

Correlation measures the **strength and direction of a linear relationship** between two variables.

- **Positive:** both rise together (study hours and marks).
- **Negative:** one rises as the other falls (absences and marks).
- **Zero:** no linear relationship.

Karl Pearson's coefficient of correlation:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \cdot \sum (y - \bar{y})^2}}$$

Value of r	Interpretation
$+1$	Perfect positive correlation
0.7 **to** 1	Strong positive
0.3 **to** 0.7	Moderate positive
0	No linear correlation
-1 **to** 0	Negative, with the same strength scale
-1	Perfect negative correlation

Spearman's rank correlation is used for ranked or ordinal data:

$$r_s = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

where d is the difference between the two ranks of each item.

> **Watch Out** **Correlation does not mean causation.** Ice-cream sales and drowning both rise in summer, but one does not cause the other. A hidden (confounding) variable, here temperature, may explain both.

3.2 Regression

Regression finds the **best-fitting line** to *predict* the dependent variable y from the independent variable x.

$$\hat{y} = a + bx \qquad b = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2} \qquad a = \bar{y} - b\bar{x}$$

- b is the **slope**: change in y for a one-unit increase in x.
- a is the **intercept**: value of y when $x = 0$.

Worked Example

x (study hours): 1, 2, 3, 4, 5. y (score): 2, 4, 5, 4, 5. So $\bar{x} = 3$, $\bar{y} = 4$.

- $\sum(x-\bar{x})(y-\bar{y}) = 4 + 0 + 0 + 0 + 2 = 6$
- $\sum(x-\bar{x})^2 = 10, \quad \sum(y-\bar{y})^2 = 6$
- $b = 6/10 = 0.6, \quad a = 4 - 0.6 \times 3 = 2.2$
- Regression line: $\hat{y} = 2.2 + 0.6x$. For $x = 6$, $\hat{y} = 5.8$.
- $r = 6/\sqrt{10 \times 6} \approx 0.77$ (strong positive)

	Correlation	Regression
Purpose	Measures strength of relationship	Predicts one variable from another
Variables	Treated equally	Independent and dependent
Result	A coefficient r	An equation

> **Note** The square of the correlation, r^2 (coefficient of determination), shows the proportion of variation in y explained by x. Here $r^2 = 0.6$, so about 60% is explained.

4 Estimation of Population Parameters

We rarely know the true population value (parameter), so we **estimate it from a sample**.

4.1 Point and Interval Estimation

- **Point estimate:** a single value, such as using the sample mean $\bar{x}$ to estimate μ.
- **Interval estimate (confidence interval):** a range likely to contain the parameter.

4.2 Properties of a Good Estimator

- **Unbiased:** its average value equals the true parameter.
- **Consistent:** gets closer to the parameter as sample size grows.
- **Efficient:** has the smallest variance among unbiased estimators.
- **Sufficient:** uses all the information in the sample.

4.3 Standard Error and Confidence Interval

The **standard error** of the mean measures how much sample means vary:

$$SE = \frac{s}{\sqrt{n}}$$

Confidence interval for the mean:

$$\bar{x} \pm (\text{critical value}) \times SE$$

Confidence level	z value	Meaning
90%	1.645	90 of 100 such intervals contain μ
95%	1.96	95 of 100 such intervals contain μ
99%	2.576	99 of 100 such intervals contain μ

For small samples ($n < 30$) with unknown population SD, use the t **value** (with $n - 1$ degrees of freedom) instead of z.

Example. $\bar{x} = 50$, $s = 10$, $n = 100$. Then $SE = 10/10 = 1$. The 95% CI is $50 \pm 1.96 \times 1 = (48.04,\ 51.96)$.

> **Watch Out** A 95% confidence interval does **not** mean there is a 95% chance that μ lies in this one interval. It means that the *method* captures the true value in 95% of repeated samples.

> **Note** A larger sample gives a smaller SE and a **narrower interval**; a higher confidence level gives a **wider interval**.

5 Hypothesis Testing

5.1 Basics

Hypothesis testing uses sample data to decide whether a claim about a population is supported.

- H_0 (null): no effect or no difference. H_1 (alternative): there is an effect or difference.
- **Significance level (α):** the risk we accept of wrongly rejecting H_0 (usually 0.05).
- **p-value:** the probability of getting such a result if H_0 were true. If $p \leq \alpha$, reject H_0.
- **One-tailed** test: direction predicted. **Two-tailed** test: any difference.

	H_0 true	H_0 false
Reject H_0	Type I error (α)	Correct decision
Do not reject H_0	Correct decision	Type II error (β)

Steps

1. State H_0 and H_1.
2. Choose the significance level α.
3. Choose the suitable test and calculate the test statistic.
4. Find the critical value (or p-value).
5. Decide: reject or fail to reject H_0, and state the conclusion in words.

> **Exam Tip** Say "**fail to reject** H_0", not "accept H_0". A test can show evidence against H_0, but it cannot prove H_0 true.

5.2 The t-test

Used to compare means when the **sample is small** ($n < 30$) or the population SD is unknown. Assumes roughly normal data.

Type	Use	Test statistic
One-sample	Compare sample mean with a known value μ_0	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad \text{df} = n - 1$
Independent two-sample	Compare means of two separate groups	Difference in means divided by its standard error
Paired	Compare before and after for the same people	One-sample t on the differences

Example (one-sample). A teacher claims the mean score is 50. A sample of $n = 36$ gives $\bar{x} = 52$, $s = 6$.

- $H_0 : \mu = 50$; $\quad H_1 : \mu \neq 50$; $\quad \alpha = 0.05$ (two-tailed).
- $SE = 6/\sqrt{36} = 1$, so $t = (52 - 50)/1 = 2.0$, with df $= 35$.
- Critical value ≈ 2.03. Since $2.0 < 2.03$, we **fail to reject** H_0: there is not enough evidence that the mean differs from 50.

5.3 Chi-square Test (χ^2)

Used for **categorical (count) data**. It compares observed frequencies (O) with expected frequencies (E).

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

- **Goodness of fit:** do observed counts match an expected pattern? df $= k - 1$.
- **Test of independence:** are two categorical variables related? df $= (r-1)(c-1)$, and $E = \dfrac{\text{row total} \times \text{column total}}{\text{grand total}}$.
- Requires counts (not percentages), and each expected frequency should generally be at least 5.

Example (goodness of fit). A die is rolled 60 times. Observed: 8, 9, 10, 11, 12, 10. Expected for each face $= 10$.

- $\chi^2 = (4+1+0+1+4+0)/10 = 1.0$, with df $= 5$.
- Critical value at 0.05 is 11.07. Since $1.0 < 11.07$, we **fail to reject** H_0: the die appears fair.

5.4 ANOVA (Concept)

Analysis of Variance tests whether the **means of three or more groups** differ.

- **Why not many t-tests?** Repeating t-tests inflates the chance of a Type I error. ANOVA tests all groups at once.
- **Idea:** compare the variation *between* group means with the variation *within* groups.

$$F = \frac{\text{Variance between groups (MSB)}}{\text{Variance within groups (MSW)}}$$

- A **large** F means group means differ more than chance would explain, so H_0 is rejected.
- H_0: all group means are equal. H_1: at least one mean is different.
- **One-way ANOVA:** one factor (e.g. teaching method with 3 types). **Two-way ANOVA:** two factors.
- **Assumptions:** independent observations, roughly normal data, equal variances across groups.
- If F is significant, a **post-hoc test** (e.g. Tukey) shows *which* groups differ.

5.5 Choosing the Right Test

Situation	Test
Compare one mean with a known value	One-sample t-test
Compare means of two independent groups	Independent t-test
Compare before and after, same people	Paired t-test
Compare means of three or more groups	ANOVA
Relationship between two categorical variables	Chi-square independence
Observed vs. expected counts	Chi-square goodness of fit
Strength of linear relation, two numeric variables	Correlation

6 Quick Revision Sheet

- **Classification** groups data; **tabulation** presents it in rows and columns.
- **Central tendency:** mean (average), median (middle), mode (most frequent).
- **Dispersion:** range, variance, SD $= \sqrt{\text{variance}}$; sample uses $n - 1$.
- **Correlation** r ranges from -1 to $+1$; correlation $\neq$ causation.
- **Regression:** $\hat{y} = a + bx$, used for prediction.
- **Estimation:** point estimate vs. confidence interval; $SE = s/\sqrt{n}$.
- **Hypothesis test:** reject H_0 if $p \leq \alpha$; Type I error is α.
- **t-test:** compare means (one sample, two sample, paired).
- **Chi-square:** categorical counts, $\sum(O - E)^2/E$.
- **ANOVA:** compare 3 or more means using F = between / within variance.

End of Module 4